

Statistics Mechanics Notes

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1 Information theory

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2 Cluster Expansion

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The cluster expansion is an approach used to construct a basis set. More precisely, it is discrete Fourier transformation on the configuration space. On the basis, we can expand any lattice dependent only functions. It's a powerful tool for the investigation of atomic ordering.

Assuming we have a configuration space, Ω . The single site in the configuration space is denoted as σ_i , and the configuration space is $(\sigma_1, \sigma_2, \dots, \sigma_N)$. For each site, it has different occupation element, that is $A = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$.

To define the orthonormal basis here, we have to give out the definition of the inner product first. The inner product on the single site σ_i is defined as:

$$\langle f|g \rangle = \sum_{a=1}^n \nu_a f(a)g(a) \quad (2.1)$$

Here ν is a reference distribution, which gives the probability of each specific element occupy this site. The orthonormal basis is a set of basis that satisfy:

$$\langle \phi_m | \phi_n \rangle = \delta_{mn} \quad (2.2)$$

It can be proved that, the number of the basis function is equal to the total number of the species. The first basis function will always be constant:

$$\phi_0 = 1 \quad (2.3)$$

The many body basis is defined as:

$$\Psi_{S,r}(\sigma) = \prod_{i \in S} \phi_{r_i}(\sigma_i) \quad (2.4)$$

S is the set of the site index in this many body term. $r_i \in \{1, 2, \dots, n-1\}$.